

GROCERY SHOP BILLING SYSTEM

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Introduction

This project presents the design and implementation of a Grocery Shop Billing System using Java and Object-Oriented Programming (OOP) principles with a Graphical User Interface (GUI). The system is designed to help small grocery shops manage billing efficiently and reduce manual calculation errors. The application allows users to add product details, calculate totals automatically, and balance for customers through a user-friendly interface. The system demonstrates core OOP concepts such as encapsulation, inheritance, abstraction, and polymorphism while using event-driven programming with Java.

Manual billing in small grocery shops can lead to errors, time delays, and inefficiency. A computerized billing system simplifies the process by automatically calculating totals and organizing product information. This project aims to develop a simple but effective billing system that enables grocery shop owners to quickly create bills and manage product purchases using a graphical interface.

Objectives

- To design a simple grocery billing system using Java.
- To apply Object-Oriented Programming concepts in a real-world application.
- To develop a user-friendly graphical interface using Java.
- To automate bill calculation and reduce human errors.
- To demonstrate event handling using ActionListener.

Scope

- The Grocery Shop Billing System allows the user to input product name, price, and quantity.
- The system then calculates the total cost for each item, balance and the overall bill.
- The system focuses on small-scale grocery stores and basic billing functionality.
- Future improvements can include database integration, inventory management, and receipt printing.

System Design

Main Classes:

- Product Class – Stores product details such as product name, price, and quantity.
- Bill Class – Handles bill generation and total calculation and balance.
- Customer Class – Stores customer details.
- Billing System Class – Controls the system operations and GUI.

Relationships:

- A Bill contains multiple Products.
- The Billing System manages product entries and bill calculations.

OOP Concepts

Encapsulation:

- Data members are declared private and accessed using getter and setter methods.

Polymorphism:

- Methods can be overloaded for different billing calculations.

Advantages

- Faster billing process
- Reduces calculation errors
- Easy to use interface
- Demonstrates practical use of OOP concepts

Future Enhancements

- Database integration for product storage
- Inventory management system
- Receipt printing
- Barcode scanning
- Customer management system

GUI Interface Description

- JFrame – Main application window
- JLabel – Display text labels
- JTextField – Input fields for product details
- JButton – Buttons for actions such as adding items and generating bills
- JTextArea – Displays the generated bill summary

Product Name : **Price :**

Quantity : **Paid Amount :**

Add Item **Generate Bill** **Clear**

----- Grocery Store -----				
Name	Item Price	Quantity	Amount	Current Total
Olive Oil	2200.0	1	2200.0	2200.0
KitKat	225.0	4	900.0	3100.0
Cheese	1350.0	1	1350.0	4450.0
Pasta	975.0	1	975.0	5425.0
Coffee	195.0	2	390.0	5815.0
Egg Pack	380.0	2	760.0	6575.0

Total = 6575.0
Paid = 10000.0
Balance = 3425.0

-- Thank you for shopping with us! --
-- Come Again --